PROJECT REPORT

PROJECT TITLE:
Spam Email Detection using Machine Learning

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🟦 1. Introduction

Spam emails are unwanted messages that often include advertisements, scams, or phishing attempts. With the rapid growth of digital communication, managing such emails manually has become inefficient. This project focuses on building a Machine Learning model that can automatically classify messages as **Spam** or **Not Spam (Ham)**.

🟦 2. Problem Statement

The goal is to develop an automated system that:

- Analyzes text messages
- Identifies whether they are spam or not
- Reduces manual effort and increases efficiency

🟦 3. Objectives

- To preprocess text data
- To convert text into numerical features
- To train a Machine Learning model
- To evaluate model performance

🟦 4. Dataset

- Dataset used: SMS Spam Collection Dataset
- It contains:
  - label → spam or ham
  - message → text content

🟦 5. Methodology

Step 1: Data Preprocessing

- Convert text to lowercase
- Remove unnecessary words
- Clean the text

Step 2: Feature Extraction

- Convert text into numerical format using **TF-IDF Vectorization**

Step 3: Model Selection

- Multinomial Naive Bayes algorithm

Step 4: Training & Testing

- Data split:
  - 80% training
  - 20% testing

🟦 6. Implementation

- Programming Language: Python
- Libraries Used:
  - Pandas
  - Scikit-learn

🟦 7. Results

- The model achieved high accuracy (around 95%)
- It successfully classifies spam and non-spam messages

🟦 8. Challenges

- Cleaning and processing text data
- Selecting appropriate features
- Avoiding overfitting

🟦 9. Conclusion

This project demonstrates that Machine Learning can effectively detect spam messages. The model performs well and can be further improved using advanced techniques.

🟦 10. Learning Outcomes

- Understanding of text preprocessing
- Knowledge of feature extraction techniques

- Experience with Machine Learning models
- Application of AI to real-world problems